

r/energy

AI infrastructure seems to run into an electricity problem pretty quickly

Posted by u/Lonely-Ad-3123 • • 0 points • 10 comments

The more I read about large AI data centers, the more it feels like electricity is the real limiting factor.

Not just generating power, but distributing it and managing heat. Apparently huge amounts of energy end up as heat somewhere in the electrical chain before it even reaches the servers.

Cooling infrastructure alone looks enormous.

That got me wondering how much work is happening on improving the efficiency of power conversion systems in these facilities.

Anyone here involved with data center infrastructure?

Comments

u/knuthf • 1 points • 2026-03-10 13:51 UTC

Wasted energy in the form of heat is caused by the use of the wrong technology.

The main problem is that business school graduates failed maths and science at secondary school level, yet now they are trying to run the world. Those who make new technology that does not produce heat are penniless, while those who provide the funds are not.

Use the wasted heat to warm houses.

u/rosier9 • 0 points • 2026-03-10 15:57 UTC

I imagine those business school graduates pointed out that summer exists, so you'll need to build chillers anyway...rendering the wildly expensive district heating redundant.

That also explains how those business school graduates ended up running businesses.

u/knuthf • 1 points • 2026-03-10 22:50 UTC

Most business school graduates cause major financial disasters. Only a few end up on Wall Street — I am one of them. Business success is rare.

u/IntelArtiGen • 1 points • 2026-03-10 12:19 UTC

> Apparently huge amounts of energy end up as heat somewhere in the electrical chain before it even reaches the servers.

It depends on many things but it should probably not be more than 10-15%, which is a lot but you can't be 100% efficient anyway in real-life conditions. Then when it's used in datacenters 100% of it ends up as heat and this is however a big problem as they have to remove this heat from the datacenters.

You can see the power supply efficiency here: https://fr.wikipedia.org/wiki/80_PLUS , the best systems (which are surely the ones they use as it doesn't cost much more compared to the cost of electricity) are >90% efficient. And for most grids in high-tech countries you have 5% losses.

u/knuthf • 1 points • 2026-03-10 14:04 UTC

Put simply, nothing rises instantly. The electricity that runs in clock cycles of 2 GHz is turned on and off at 2 GHz. So, 2,000,000,000 is the number of times it turns on per second, going from 0 to 1. The rise follows a sine curve, and the difference between the sine curve and the square is heat. In metals, it is resistance. You can cool the chips down to absolute zero and achieve superconductivity.

The Chinese have moved on to carbon nanotubes and graphene, using coal to make molecules lock into one another to create superconductors. They are also testing optical circuits. These will not generate heat they have no resistance - along, and isolate across. However, this is advanced physics that business school graduates do not understand. In the old days, we had to make the leap from TTL to CMOS. I have made that leap.

u/iwriteaboutthings • 1 points • 2026-03-10 11:51 UTC

I am sure there are many smart things people will do to make this a bit more efficient, but the current issue (economically) is that it takes longer to build power infrastructure than build AI infrastructure and the surge of demand for new power came as a surprise. As a result the grid is "behind," but there is nothing fundamentally difficult about serving the demand.

The other aspect is society likes to think of power as an always available basic product. Power being unavailable feels like a failure of your utility. GPUs being unavailable is "wow, NVidia is king of the world!"

u/Quietly_here_28 • 1 points • 2026-03-10 09:37 UTC

Hillcrest Energy is one of the companies working on inverter technology aimed at reducing switching losses. Their focus is improving efficiency in systems like EV drivetrains and grid electronics. It's still a

pretty technical niche though.

u/Additional_Clothes58 • 2 points • 2026-03-10 08:49 UTC

"Apparently huge amounts of energy end up as heat somewhere in the electrical chain before it even reaches the servers."

this is entirely untrue. transmission losses are 10-15%. the rest of it is electricity delivered to the data centre where 30-40% goes in running the cooling systems and the rest in running the servers themselves. Obviously, the entire electricity consumed by the servers ultimately ends up as heat.

u/PeakAccomplished2431 • 1 points • 2026-03-10 08:30 UTC

A friend of mine works for a company that installs electrical systems in large buildings. He said one of the constant headaches is heat. The bigger the system gets, the harder it becomes to manage.

u/buttershutter69 • 1 points • 2026-03-10 05:49 UTC

Power conditioning and power delivery are a huge part of large facilities. UPS systems, voltage control, converters. There is a lot going on behind the scenes before electricity ever reaches a rack of servers.